Study Helper App: v1 spec

Handoff for the build chat

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Summary. v1 is one Expo (React Native) codebase shipped as an Android APK and a web link with login, on Supabase. It schedules study around fixed life, logs real time, notifies on Android, and keeps Claude in the loop through zip export and CSV import. Build starts after the Fuzzy exam (Wed 28 Oct). Another Claude chat writes the code, Sergio reviews.

Rules for the build chat

- Never use em dashes anywhere: UI text, code comments, docs, commit messages, replies.
- Reply to Sergio in short caveman style: fragments, little filler, keep key explanations.
- He prefers documents in LaTeX (with a compiled PDF next to the `.tex`).
- He still has Cloud (Wed 4 Nov) and Deep Learning (Thu 19 Nov). Batch questions, keep reviews short.

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1 Goal

Replace the current Study Helper tracker (a Claude artifact with a small db, plus local CSV/JSON exports) with a real app.

- All-day study schedule helper. Juggles exams, projects, homework, labs, quizzes and lecture catch-up in the free time around fixed life (classes, gym, Muay Thai, commute, sleep).
- Later: grows into a full life schedule. Events stay generic enough for that, but v1 does not build it.
- Single user now, multi-user ready: per-user data from day 1, no sharing features.
- Claude stays in the loop for deep plan refines: app exports a zip, Claude returns CSV, app imports it. v2: Claude reads/writes the cloud db directly.
- Current tracker stays live until the app passes the migration check (Section 14).

2 Platform and stack

Part	Choice
App	Expo (React Native) + Expo Router, TypeScript. One codebase for Android + web (react-native-web).
Backend	Supabase: Postgres, Auth, Row Level Security on every table. Edge Functions only in v2.
Auth	Email + password (or magic link). No Google login (user preference).
Web	Static export (`expo export -p web`) on Netlify or Cloudflare Pages (free). Gives the link.
Android	EAS build, `preview` profile, APK sideload. No Play Store.
State	TanStack Query (server state, persisted cache) + Zustand (UI and timer state).
UI	Builder's pick, suggest NativeWind. Dark + light theme.
Notifications	`expo-notifications` local scheduled notifications on Android. Web: none in v1 (in-page sound while tab open only).
Time	Timestamps stored in UTC. Planning and display in user timezone (default Asia/Beirut).

Part	Choice
Offline	Android works offline for today's plan, timer and ticking. Writes go to an outbox queue, sync on reconnect, last-write-wins per row by `updated_at`.

Suggested repo layout: `app/` (screens), `src/scheduler/` (pure scheduling logic, no React, unit tested), `src/io/` (export zip, CSV import, migration), `supabase/migrations/`.

3 Data model

Every table has `id uuid`, `user_id uuid` (RLS: `user_id = auth.uid()`), `created_at`, `updated_at`, `archived_at`. A course holds items, an item holds ordered chunks, and blocks put chunks into time.

Table	Key fields
`courses`	`code` (CPEN448), `name`, `short` (Cloud), `color`, `professor`, `semester`, `status` active/archived, `notes`
`items`	`course_id`, `type` exam / project / homework / lab / quiz / catchup, `title`, `due_at`, `due_confirmed` (exam slot confirmed vs assumed), `scope`, `rules` (cheat sheet rules etc.), `needs_laptop`, `weight_pct`, `priority_boost` (default 0), `status`
`chunks`	`item_id`, `order`, `title` (L3 p19-33), `kind` notes / problems / review / mock / cheat_sheet / lab_work / writing / other, `source_ref` (file + pages), `est_blocks`, `done_blocks`, `milestone_due` (project stages), `status` todo/partial/done/skipped
`fixed_events`	`label`, `kind` class / sport / commute / sleep / other, `weekdays`, `start_time`, `end_time`, `valid_from`, `valid_to`, `never_cut` (true for gym and Muay Thai), `course_id` (for classes)
`oneoff_blocks`	`label`, `date`, `start_time`/`end_time` or `all_day`, `kind` blocked / off / trip
`blocks`	`date`, `start`, `end`, `course_id`, `item_id`, `chunk_ids`, `what`, `type` Notes / Problems / Review / Mock / CheatSheet / Buffer / Rest / Break / Fixed, `source` auto / claude / manual / import, `locked`, `status` planned / done / partial / skipped / moved, `subtasks [{name, plan_min}]`, `legacy_key` (old keys like `1016-1835-Opt`), `import_batch_id`
`time_logs`	`block_id`, `started_at`, `ended_at`, `plan_min`, `actual_min`, `pages`, `completed`, `focus` 1-5 (optional), `note`, `subtasks [{name, plan, act}]`
`milestones`	`date`, `title`, `parts`, `done` (keeps the current daily milestone feature)
`active_timer`	One row per user so web and phone show the same timer: `block_id`, `phase` study/break, `phase_started_at`, `phase_ends_at`, `subtask_index`, `auto`
`settings`	One row per user (jsonb is fine): rules below, notification toggles, quiet hours, timezone
`import_batches`	`source` claude_csv / migration / zip, `summary`; used for undo

Default rules (settings)

- Pomodoro 50 study / 10 break, long break 20-30 min after every 3 blocks. Blocks can be any length (25, 40, 75 min mocks); 50/10 is only the default.

- Switch subject after 2 blocks of the same course, except exam-eve review days.
- Hard stop 23:00, wake 08:20, daily review block 22:00-22:30.
- First block after each exam = rest. Sunday afternoon/evening partly off.
- Fresh slots (mornings, first block after lunch) go to the highest pressure item.
- Gym and Muay Thai are `never_cut`: necessary breaks.

4 Screens

1. **Login**
2. **Today** (home): timeline of the day, fixed events greyed, study blocks in course colors. Current block highlighted with a big Start button, subtasks list, next up, behind/ahead indicator.
3. **Week**: 7-day view. Tap to edit, drag to move (at least on web), lock toggle.
4. **Courses**: course list, then course page (items by type with due dates and pressure), then item page (ordered chunks with est/done blocks). Add, edit, archive at every level.
5. **Quick add**: homework / lab / quiz in a few taps (course, due, est blocks, needs laptop).
6. **Archive**
7. **Stats**
8. **Import / Export**
9. **Settings**: rules, fixed schedule editor, notifications, theme, account.

5 Timer and logging

Keeps the per-subtask plan vs actual timing the current tracker already has.

- Pomodoro start / pause / stop, auto break after a block, optional auto start of the next block (`auto` flag).
- Each subtask has plan minutes; the app records actual minutes and shows the ratio.
- Timer runs from stored timestamps, so it survives app backgrounding. Phase end fires a notification.
- On block end: done / partial (which subtasks or pages) / skipped. Actual time comes from the timer and stays editable. Optional short note and optional focus 1-5.
- Partial or skipped work flows back to the chunk's `done_blocks`, so the scheduler knows what is left.

6 Scheduler (hybrid)

The app auto-fills and repairs day to day. Claude does deep refines through CSV. Implement as a pure deterministic function `schedule(input) -> { blocks, warnings }` in `src/scheduler/`, with unit tests.

Pressure score

For each active item:

$$\text{pressure} = \frac{\text{remaining estimated blocks}}{\max(0.5,\ \text{days until due})} + \text{priority_boost}$$

Days until due are fractional (to `due_at`). Items with no due date (lecture catch-up) get a low default. Highest pressure gets the most blocks and the freshest slots.

Behaviour

- **Auto-fill**: fills free slots for the next 7 days (configurable), respecting fixed events, one-offs, hard stop, pomodoro lengths, the 2-then-switch rule, rest after exam, Sunday off time and the daily review. Chunks are taken in order. Locked blocks and Claude blocks are never moved unless the user allows it.
- **Missed blocks**: block ended without a tick triggers a nag, then: done / partial / move to nearest buffer / let auto-fill reschedule. Nearest exam first when picking buffers.
- **Rebalance** button: recompute, show a diff (added / moved / removed), accept all or pick.
- **Overload**: if needed blocks exceed free slots before a due date, show a red banner, send a warning, and suggest what to shrink.

7 Notifications (Android, v1)

All toggleable in settings, all respecting quiet hours (after hard stop, during classes, gym, Muay Thai).

- Block start, block end, break end.
- Morning summary at wake time: today's blocks + top pressure item.
- Deadline warnings 7 / 3 / 1 days before each due date (configurable), plus overload warning.
- Nag 10 min after a block ends with no tick.

Local notifications are rescheduled for the next 48 h whenever the plan changes or the app opens.

8 Archive and stats

Archive

- Done blocks and done items per course, searchable, filtered by course / date / type, with notes and time logs.
- Finished courses and old semesters: archiving a course moves it and all its items out of active views.
- Restore anything back to active.

Stats

- Hours per course per week.
- Estimate vs actual ratio per course and per block type (like the current Ratio column).
- Planned vs done, streaks, focus trend.

9 Claude loop: export zip and CSV import

9.1 Export zip

File name `study-helper-export_YYYY-MM-DD_HHMM.zip`. Range: all / last 7 days / since last export. Android share sheet or web download.

- `manifest.json`: schema version, exported_at, timezone, range, counts.
- `README.md`: short guide for Claude: what each file is, the import format below, the rules (no em dashes, keep gym/Muay Thai, 2-then-switch, hard stop).
- `courses.csv`, `items.csv`, `chunks.csv`, `blocks.csv`, `time_logs.csv`, `time_log_subtasks.csv`, `fixed_events.csv`, `oneoff_blocks.csv`, `milestones.csv`, `settings.json`.
- `pressure.csv`: current pressure score per item, so Claude sees what the app sees.
- `snapshot.json`: everything in one file.

9.2 CSV import (what Claude sends back)

`blocks_import.csv`:

Column	Notes
`op`	add / update / delete
`id`	blank for add, block uuid for update/delete
`legacy_key`	optional, matches old tracker keys
`date`	YYYY-MM-DD
`start, end`	HH:MM
`course`	course short (Opt, PM, Fuzzy, Cloud, DL)
`item`	item title or id
`chunks`	chunk titles separated by ; (optional)
`what`	block text
`type`	Notes / Problems / Review / Mock / CheatSheet / Buffer / Rest / Break
`subtasks`	`name|min; name|min`
`locked`	true/false, default true for Claude blocks

Also `chunks_import.csv` (`op, id, course, item, order, title, kind, source_ref, est_blocks, milestone_due`) and `items_import.csv` (`op, id, course, type, title, due_at, due_confirmed, scope, needs_laptop`).

Flow: pick files, validate (row-level errors), diff preview, apply all or pick rows, saved as an import batch, **Undo last import** button.

10 Migration from the current tracker

Inputs live in `D:\Uni\5th year\Fall Semester\Tracker`. They are stale (snapshot 20 Oct, blocks CSV 25 Oct). Before migrating, Sergio asks Claude to dump the live tracker db into a fresh `Tracker snapshot.json` in the same format. The importer must take that format.

Source	Shape	Maps to
`meta/plan.data.rows[]`	`k` key, `w` week, `d` date, `s` start, `e` end, `sub` subject, `t` what, `ty` type	blocks (`legacy_key = k`); Fixed rows become fixed events or one-offs; Break/Rest become break blocks
`progress/state.data.blocks`	`{k: true}`	block status done
`plans/<k>.data.subs[]`	`{n, m}`	block subtasks (name, plan_min)
`timelog/<k>.data`	`plan, act, pages, completed, startedAt, endedAt, subs[{n, plan, act}]`	time_logs
`Tracker_Blocks.csv`	Week, Day, Start, End, Subject, What, Type, Done	blocks (fallback)
`Time_Log_*.csv`	Block, Subject, Type, Started, Ended, Plan_min, Actual_min, Ratio, Pages, Completed, Subtask, Sub_plan, Sub_actual, Sub_ratio (subtask rows follow their block row)	time_logs (fallback)

Courses are created from subject shorts (Opt, PM, Fuzzy, Cloud, DL); codes filled in later. `timer/active` is ignored.

11 Seed data (Fall 2026)

- **Courses**: Optimization Theory ELEN401 (Opt), Project Management GENG402 (PM), Fuzzy Logic Control ELEN527 (Fuzzy), Cloud Computing and Big Data CPEN448 (Cloud), Deep Learning CPEN426 (DL). Graduation Project GENG490 too. Seminar is class only.
- **Exams**: Opt Thu 22 Oct (done), PM Mon 26 Oct (done), Fuzzy Wed 28 Oct 9:30, Cloud Wed 4 Nov 11:00 (assumed slot), DL Thu 19 Nov (comprehensive).
- **Classes**: Mon/Wed 9:30-14:00, Tue 9:30-12:30, Thu 9:30-14:00. None Fri to Sun. Wake 8:20, leave 8:50, commute 30 min each way.
- **Sport**: Muay Thai Mon/Wed/Fri 19:00-20:30, blocked 18:40-21:20 with travel and shower. Gym Tue 13:00-14:30, Sat 12:30 (about 1.5 h incl. shower), maybe Thu. Both never cut.
- **Study method**: 50/10 Pomodoro, one task per block, examples first then theory, handwritten notes, switch subject every 2 blocks, rest block after each exam, hard stop 23:00.

12 Out of scope for v1 (v2 list)

- Web push notifications (Supabase cron + Web Push).
- Claude direct access to the cloud db (small MCP server or scoped API over Supabase).
- Calendar: no Google. v2 option is a read-only `.ics` feed any calendar app can subscribe to.
- Full life schedule (non-study tasks, habits, chores).

- Course material uploads (slides/PDFs per course for Claude context).
- Sharing and multi-user onboarding. iOS.

13 Build order

M3 is the switch-over point: once the Claude loop works end to end, the old tracker can retire.

#	Milestone	Done when
M0	Expo + TS + Router, Supabase schema + RLS, auth, web deploy link, first APK	Login works on phone and web
M1	Courses / items / chunks CRUD, fixed events, one-offs, settings	Cloud + DL entered with chunks
M2	Today + Week views, manual blocks, tick + log, timer with subtasks, shared active timer	Daily use matches the current tracker
M3	Migration importer, export zip, CSV import with diff + undo	Claude loop works end to end (**switch-over**)
M4	Scheduler (pure fn + tests), auto-fill, rebalance diff, missed-block flow, overload warning	Auto week plan passes tests
M5	Android notifications	All 4 types fire with app closed
M6	Archive + stats	Archive, restore and stats screens work
M7	Offline outbox, polish	A full day in airplane mode works

14 Acceptance checks

- ☐ Migration: imported plan matches the tracker for 25 and 26 Oct; ticked count and subtasks match the fresh snapshot.
- ☐ Export then re-import gives identical data (round trip).
- ☐ Scheduler: no overlap with fixed events or one-offs, nothing after 23:00, gym/Muay Thai never cut, 2-then-switch kept except exam eve, rest after each exam, locked and Claude blocks untouched, deterministic output.
- ☐ Notifications fire on Android with the app closed and phone locked.
- ☐ RLS: a second test user cannot read the first user's rows.
- ☐ Lint/grep check fails the build if any source or UI string contains an em dash.